

1. Administrative & Study Identification

Full Study Title: A Phase III, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of Astegolimab in Patients With Chronic Obstructive Pulmonary Disease **Short Title/Acronym:** ARNASA **Protocol Number:** GB44332 **NCT Number:** NCT05595642 **Sponsor Name:** Hoffmann-La Roche (Genentech) **Investigational Product:** Astegolimab (an investigational, fully human anti-ST2 monoclonal antibody) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of astegolimab compared with placebo in reducing the annualized rate of moderate and severe COPD exacerbations over a 52-week treatment period. **Secondary Objectives:** To evaluate the time to the first moderate or severe COPD exacerbation, patient-reported outcomes (St. George's Respiratory Questionnaire-COPD [SGRQ-C], Evaluating Respiratory Symptoms in COPD), and change from baseline in post-bronchodilator FEV1 at Week 52.

2.2 Background & Rationale To address the medical necessity for new treatment options to reduce exacerbation rates in patients with moderate to severe COPD. Exacerbations are linked to the interleukin (IL)-33/ST2 pathway. Astegolimab is designed to block the ST2 receptor, inhibiting IL-33 signaling, thereby attempting to lower the number of moderate and severe exacerbations in patients who are former or current smokers with a history of frequent flare-ups.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (Astegolimab SC Q2W, Astegolimab SC Q4W, Placebo Q2W)

3.2 Planned Enrollment & Duration Target \$N\$: ~1,375 patients (1,290 originally planned, 1,375 actually enrolled) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 29-Dec-2022 **Estimated Primary Completion:** 24-Jun-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age 40–80 years. 2. Former or current smokers with a COPD diagnosis ≥ 12 months prior to screening. 3. Post-bronchodilator percent predicted FEV1 $\geq 20\%$ and $< 80\%$, and FEV1/FVC < 0.7 at Visit 1 or 2. 4. Experienced ≥ 2 moderate or severe COPD exacerbations within the 12 months prior to screening.

4.2 Exclusion Criteria 1. Current diagnosis of asthma or other significant respiratory disease. 2. NYHA Class III/IV heart failure. 3. Patients receiving palliative treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Astegolimab administered either once every 2 weeks (Q2W) or once every 4 weeks (Q4W). To maintain blinding, participants on the Q4W arm alternate with placebo injections to match the Q2W schedule. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** 52-week treatment period, followed by a 12-week safety follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care maintenance therapy (e.g., Inhaled Corticosteroids [ICS] + Long-Acting Beta-Agonist [LABA]; Long-Acting Muscarinic Antagonist [LAMA] + LABA; or ICS + LAMA + LABA). **Prohibited:** Other investigational drugs or non-approved biological therapies outside the clinical trial setting.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Exacerbation History Confirmation, FEV1/FVC Spirometry
Baseline	Day 0	Randomization (1:1:1), First SC Injection, Baseline symptom assessments
Interim	Weeks 2 to 50	Ongoing SC injections (Q2W/Q4W), Safety monitoring, Exacerbation evaluation, Patient-Reported Outcomes (PROs)
End of Study	Week 52 & Week 64	Final efficacy assessments (Week 52), followed by a 12-week safety follow-up (Week 64) for patients not entering the Open-Label Extension

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Reduction in the annualized rate of moderate and severe COPD exacerbations (AER) from baseline over the 52-week treatment period. **Measurement Method:** Clinical assessment of acute worsening in airway function and respiratory symptoms, standardized per total patient-years.

7.2 Secondary & Exploratory Endpoints * Time to first moderate or severe COPD exacerbation. * Patient-reported outcomes (St. George's Respiratory Questionnaire-COPD [SGRQ-C] and Evaluating Respiratory Symptoms in COPD). * Change from baseline in post-bronchodilator FEV1 at Week 52.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine safety evaluations, physical examinations, and continuous tracking of respiratory symptoms and injection-site reactions. **Serious Adverse Events (SAEs):** Standard rapid reporting (typically 24 hours) for exacerbations requiring hospitalization or life-threatening events. **Data Monitoring Committee (DMC):** Independent safety monitoring panel evaluating benefit-risk profiles throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population including all randomized participants (regardless of blood eosinophil count). **Sample Size Justification:** Approximately 1,375 patients, adequately powered to detect a statistically significant reduction in annualized exacerbation rates against placebo. **Handling of Missing Data:** Standardized imputation for missing longitudinal variables and exposure-adjusted negative binomial regression models for exacerbation rates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote monitoring of trial sites to ensure adherence to standard of care maintenance and investigational product dispensing. **Audit Trail:** eCRF locking and clinical data management systems for tracking PRO questionnaires and exacerbation records.

11. Study Identifiers & Governance

Primary ID: GB44332 **Registry Number:** NCT05595642 **Sponsor/Lead Organization:** Hoffmann-La Roche **Study Title:** ARNASA **Official Regulatory Title:** A Phase III, Randomized, Double-Blind, Placebo-Controlled, Multicenter Study to Evaluate the Efficacy and Safety of Astegolimab in Patients With Chronic Obstructive Pulmonary Disease

12. Clinical Framework (COPD Specific)

Primary Condition: Chronic Obstructive Pulmonary Disease (COPD)
Diagnosis Threshold: Post-bronchodilator FEV1 $\geq 20\%$ and $< 80\%$, with FEV1/FVC < 0.7
Medical Methodology: Standard of care maintenance therapy (ICS, LABA, and/or LAMA combinations)
Optimization Target: IL-33/ST2 pathway blockade for exacerbation reduction

13. Study Procedures & Methodology

Management Pathway: Subcutaneous Astegolimab injections + background maintenance therapy
Education Protocol (HCP): Protocol training on exacerbation severity grading and PRO administration
Education Protocol (Patient): Symptom diary training and SC injection observation/reporting
Quality Audit Mechanism: Tracking patient adherence to daily/weekly respiratory symptom diaries (e-diary) and study drug schedules

14. Observation & Follow-Up Schedule

Total Duration: 52 weeks (treatment) + 12 weeks (safety follow-up)
Onsite Visit Cadence: Frequent regular visits to accommodate Q2W injections and lung function spirometry
Remote Monitoring Frequency: E-diary/questionnaire completion monitoring (Ongoing)
Checklist Review Interval: Continuous daily respiratory symptom diaries

15. Sign-off & Approval

Role	Name	Signature	Date						
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]						
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]						
Statistician	[Name]	_____	[DD-MMM-YYYY]						